HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Mathematics

F18CE

Multivariable Calculus and Real Analysis B

Semester 2 – 2018/19

Duration: 2 Hours

Attempt all questions

A University approved calculator may be used
for basic computations, but
appropriate working must be shown to obtain full credit.

1 Let (a_n) be a sequence of real numbers. Define what is meant by

$$\lim_{n\to\infty} a_n = L.$$

Using this definition, prove the following results.

(a) $\lim_{n\to\infty} \dfrac{n^3+5n+7}{n^3+3n+2} = 1\,.$

(b) If $\lim_{n\to\infty} a_n = L > 0$ then there exists an integer N such that

$$a_n < 2L, \quad \text{for all } n \geq N.$$

(c) If $\lim_{n\to\infty} a_n = L$ then $\lim_{n\to\infty} (a_{n+1} - a_n) = 0.$ **[11 marks]**

2 Give examples of sequences, one in each case, of the following.

(a) (a_n) is convergent and $(1/a_n)$ is also convergent.

(b) (a_n) and (b_n) are not convergent, but (a_n/b_n) is convergent.

(c) (a_n) does not have a convergent subsequence.

(d) (a_n) monotone but not convergent. **[8 marks]**

3 (a) Let A be a non-empty set of real numbers. Define what is meant by A is bounded above and, if A is bounded above, define the supremum of A.

(b) Let A be a non-empty, bounded set of positive real numbers. Let B be the set consisting of numbers of the form

$$b = a^2 + a + 1, \quad \text{for } a \in A.$$

Show that B is bounded, and that

$$\sup B \leqslant (\sup A)^2 + \sup A + 1.$$ **[5 marks]**

4 (a) State the Bolzano-Weierstrass Theorem.

(b) Give an example of a bounded sequence having subsequences converging to different limits.

(c) Suppose that the sequence (a_n) is defined by

$$a_n = \frac{(-1)^n n^3 + 5n + 10}{2n^3 + n + 1}, \quad n \geqslant 1.$$

Show that (a_n) has a convergent subsequence. **[6 marks]**

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5 Suppose that $f : A \to \mathbb{R}$ is a function defined on an open interval A, and $a \in A$. Define what is meant by

$$\lim_{x \to a} f(x) = L.$$

(a) Show that if x satisfies $|x + 5| < 1$ then $|x| < 6$.

(b) Using the above definition of the limit, prove that

$$\lim_{x \to -5} (x^2 + 2x) = 15. \qquad \textbf{[7 marks]}$$

6 (a) Suppose that $f : A \to \mathbb{R}$ is a function defined on an open interval A, and $a \in A$. State the limit definition of f is continuous at a.

(b) Suppose that the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are continuous at a. Show that the function $f + g : \mathbb{R} \to \mathbb{R}$ is continuous at a
(you can use standard theorems about limits here). **[5 marks]**

7 (a) State the Intermediate Value Theorem and Rolle's Theorem.

(b) Use these theorems to show that the equation

$$\cos(\pi x) = x + x^2$$

has a unique solution in $(0, 1)$. **[10 marks]**

8 (a) Suppose that $f : A \to \mathbb{R}$ is a function defined on an open interval A, and $a \in A$. State the definition of f is differentiable at a.

(b) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is defined by

$$f(x) = \begin{cases} -2x^2, & x < 0, \\ 5x^2, & x \geq 0. \end{cases}$$

Prove that f is differentiable at 0. **[5 marks]**

9 (a) State the First Mean Value Theorem.

(b) Use the First Mean Value Theorem to prove that

$$\sin x \leqslant x, \quad \text{for } x \in [0, \pi]. \qquad \textbf{[5 marks]}$$

10 (a) State the Comparison Test for series.

(b) For what values of $\alpha > 0$ does the series $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^\alpha}$ converge?

(c) Discuss the convergence of the series $\displaystyle\sum_{r=1}^{\infty} a_r$ for the following cases:

$$(i) \quad a_r = \frac{r^2 + \cos r}{r^4 + 5r^2 + 6}; \qquad (ii) \quad a_r = \frac{r^4 + 4r + 5}{r^5 + 2r^3 + 3r^2}. \qquad \textbf{[10 marks]}$$

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11 Find the radius of convergence R of the power series

$$\sum_{r=0}^{\infty} \frac{r!\, x^r}{(2r)!}.$$ **[3 marks]**

12 (a) Let $f : [a, b] \to \mathbb{R}$ be a bounded function and let $P = \{x_0, x_1, x_2, \ldots, x_k\}$ be a partition of $[a, b]$. Define the upper and lower sums $U(P, f)$ and $L(P, f)$.

(b) Let $g : [0, 3] \to \mathbb{R}$ be defined by

$$g(x) = \begin{cases} 5, & 2 \leqslant x < 3, \\ 7, & x = 3, \\ 11, & 3 < x \leqslant 5. \end{cases}$$

Let P_n be the partition given by $P_n = \left\{2, 3 - \frac{1}{n}, 3 + \frac{1}{n}, 5\right\}$.

Find the upper and lower sums $U(P_n, g)$ and $L(P_n, g)$.

Hence show that g is integrable on $[2, 5]$ and find $\displaystyle\int_2^5 g(x)\, dx$. **[10 marks]**

END OF EXAM

1 **Definition** For any $\epsilon > 0$ there exists an integer $N > 0$ such that

$$|a_n - L| < \epsilon \quad \text{for all } n \geq N.$$ (2)

(a) Let $\epsilon > 0$.

$$\left|\frac{n^3+5n+7}{n^3+3n+2} - 1\right| = \left|\frac{3n+5}{n^3+3n+2}\right| \leqslant \left|\frac{7n}{n^3}\right| < \epsilon \quad \text{if} \quad n^2 > \frac{7}{\epsilon}.$$

So if we choose N such that $N > \sqrt{7/\epsilon}$, then

$$|a_n| < \epsilon \quad \text{for all } n \geq N.$$ (3)

(b) Let $\epsilon = L$. Choose N such that

$$n \geqslant N \implies |a_n - L| < L \implies a_n - L < L \implies a_n < 2L.$$ (3)

(c) Let $\epsilon > 0$. Choose N such that $n \geqslant N \implies |a_n - L| < \frac{1}{2}\epsilon$. Then

$$n \geqslant N \implies |a_{n+1} - a_n| = |(a_{n+1} - L) - (a_n - L)| \leqslant |a_{n+1} - L| + |a_n - L| < \frac{1}{2}\epsilon + \frac{1}{2}\epsilon = \epsilon.$$ (3)

2 (a) $a_n = 1, 1, 1, \ldots$; (2)

(b) $a_n = n$, $b_n = n$, $n = 1, 2, \ldots$; (2)

(c) $a_n = n$. (2)

(d) $a_n = (-1)^n/n$. (2)

3 (a) **Definition** A set A is bounded above if there exists a real number K such that

$$a \leq K \qquad \text{for all } a \in A.$$

If A is bounded above, then the *supremum* of A is the least real number K that is an upper bound of A. It is denoted $\sup A$. (2)

(b) By the definition of $\sup A$:

$0 \leqslant a \leqslant \sup A$ for all $a \in A \implies 0 \leqslant a^2 + a + 1 \leqslant (\sup A)^2 + \sup A + 1$ for all $a \in A$.

Hence, B is bounded and $(\sup A)^2 + \sup A + 1$ is an upper bound for B, and since $\sup B$ as the least upper bound for B:

$$\sup B \leqslant (\sup A)^2 + \sup A + 1.$$ (3)

4 (a) **Theorem** [Bolzano-Weierstrass theorem (BWT)] *Every bounded sequence has a convergent subsequence.* (2)

(b) $a_n = 1, -1, 1, \ldots$; (1)

(c)

$$|a_n| \leqslant \frac{n^3+5n+10}{2n^3+n+1} \leqslant \frac{16n^3}{2n^3} \leqslant 8,$$

so (a_n) is bounded, and so by BWT has a convergent subsequence. (3)

5 **Definition** For any $\epsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{if } 0 < |x - a| < \delta.$$

(2)

(a) $|x+5| < 1 \implies -1 < x+5 < 1 \implies -6 < x < -4 \implies |x| < 6.$ (2)

(b) Let $\epsilon > 0$.

$$|x^2 + 2x - 15| = |(x+5)(x-3)| \leqslant |x+5|(|x|+3) < 9|x+5|,$$

by (a). Now let

$$\delta = \min\left\{1, \frac{\epsilon}{9}\right\}.$$

Then:

$$|x^2 + 2x - 15| < 9|x+5| < \epsilon \quad \text{if } 0 < |x+5| < \delta,$$

so that

$$\lim_{x \to -5}(x^2 + 3x) = 10.$$

(3)

6 (a) **Definition** f is continuous at a if the limit $\lim_{x \to a} f(x)$ exists, and $\lim_{x \to a} f(x) = f(a)$. (2)

(b) Since f and g are continuous, $\lim_{x \to a} f(x) = f(a)$, $\lim_{x \to a} g(x) = g(a)$, so by a standard theorem on limits

$$\lim_{x \to a}(f(x) + g(x)) = f(a) + g(a)$$

which shows that $f + g$ is continuous. (3)

7 (a) **Theorem** (IVT) *Let f be continuous on $[a, b]$, and suppose that a number γ satisfies*

$$f(a) \leqslant \gamma \leqslant f(b) \quad \textit{or} \quad f(a) \geqslant \gamma \geqslant f(b).$$

Then there exists $c \in [a, b]$ such that $f(c) = \gamma$. (2)

Theorem (Rolle) *Let f be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$ then there exists $c \in (a, b)$ such that $f'(c) = 0$.* (2)

(b) Let $f(x) = \cos(\pi x) - x - x^2$, $x \in [0, 1]$.
Then f is continuous on $[0, 1]$ and differentiable on $(0, 1)$.
Now,

$$f(0) = 1 > 0, \quad f(1) = -1 - 2 < 0,$$

so by the IVT there exists $c \in [0, 1]$ such that $f(c) = 0$, that is, $\cos(\pi c) = c + c^2$. (3)
Also,

$$f'(x) = -\sin(\pi x) - 1 - 2x < 0 \quad \text{on } (0, 1),$$

so there cannot be 2 solutions by Rolle. (3)

8 (a) **Definition** The function f is *differentiable* at $a \in \mathbb{R}$ if the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists. (2)

(b) Put $a = 0$ and $h = x - a = x$. Then the left and right hand limits are:

$$h < 0 \implies \frac{f(a+h) - f(a)}{h} = \frac{f(h)}{h} = \frac{-2h^2}{h} = -2h \to 0,$$
$$h > 0 \implies \frac{f(a+h) - f(a)}{h} = \frac{f(h)}{h} = \frac{2h^2}{h} = 2h \to 0.$$

The left and right hand limits are equal, so the limit exists and f is differentiable at 0. (3)

9 (a) **Theorem** (First Mean Value Theorem) Let f be differentiable at all points of the open interval (a, b) and continuous on $[a, b]$. Then, there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$ (2)

(b) Define $f(x) = \sin x - x$ for $x > 0$. Then f is continuous on $[0, \pi]$, and differentiable on the open interval $(0, \pi)$, and

$$f'(x) = \cos x - 1 < 0, \quad \text{for } 0 < x < \pi.$$

By the Mean Value Theorem, on the interval $[0, x]$, with $0 < x \leqslant \pi$, there exists $c \in (0, x)$ such that

$$f(x) = f(0) + f'(c)x < 0 \implies \sin x > x.$$ (3)

10 (a) **Theorem** (Comparison Test) Suppose that $\sum_{r=1}^{\infty} a_r$, $\sum_{r=1}^{\infty} b_r$ are series with

$$0 \leqslant a_r \leqslant b_r, \quad \text{for all } r \geqslant 1.$$

Then:

(i) $\sum_{r=1}^{\infty} b_r$ converges $\implies$ $\sum_{r=1}^{\infty} a_r$ converges;

(ii) $\sum_{r=1}^{\infty} a_r$ diverges $\implies$ $\sum_{r=1}^{\infty} b_r$ diverges. (2)

(b) The given series converges if $\alpha > 1$ (and diverges if $\alpha \leqslant 1$). (2)

(c) (i) For $r \geqslant 1$, $a_r = \dfrac{r^2 + \cos r}{r^4 + 5r^2 + 6} \leqslant \dfrac{2r^2}{r^4} = 2\dfrac{1}{r^2}$

(making the top line bigger and the bottom line smaller)
and the series $1/r^2$ converges, so the original series converges. (3)

(ii) For $r \geqslant 1$, $a_r = \dfrac{r^4 + 4r + 5}{r^5 + 2r^3 + 3r^2} \geqslant \dfrac{r^4}{6r^5} = \dfrac{1}{6r}$

(making the top line smaller and the bottom line bigger)
and the series $\sum 1/r$ diverges, so the original series diverges.

(3)

11 By the Ratio test,

$$\frac{(r+1)!}{(2r+2)!}\frac{(2r)!}{r!}|x| = \frac{r+1}{(2r+2)(2r+1)}|x| \to 0 \qquad \text{for any } x,$$

so the power series converges for x, so the radius of convergence is $R = \infty$. (3)

12 (a) Upper and lower sums:

$$U(P,f)=\sum_{r=0}^{k-1} S_r(x_{r+1}-x_r), \qquad S_r=\sup\{f(x):x\in[x_r,x_{r+1}]\},$$

$$L(P,f)=\sum_{r=0}^{k-1} I_r(x_{r+1}-x_r), \qquad I_r=\inf\{f(x):x\in[x_r,x_{r+1}]\}. \qquad \textbf{2}$$

(b) For the given function g and partitions P_n:

$$\sup\left\{f(x):x\in\left[2,3-\frac{1}{n}\right]\right\}=5\,,$$

$$\sup\left\{f(x):x\in\left[3-\frac{1}{n},3+\frac{1}{n}\right]\right\}=11\,,$$

$$\sup\left\{f(x):x\in\left[3+\frac{1}{n},5\right]\right\}=11$$

so that

$$U(P_n,f)=5\left(\left(3-\frac{1}{n}\right)-2\right)+11\left(3+\frac{1}{n}-\left(3-\frac{1}{n}\right)\right)+11\left(5-\left(3+\frac{1}{n}\right)\right)=27+\frac{6}{n}\cdot$$

Similarly,

$$L(P_n,f)=5\left(\left(3-\frac{1}{n}\right)-2\right)+5\left(3+\frac{1}{n}-\left(3-\frac{1}{n}\right)\right)+11\left(5-\left(3+\frac{1}{n}\right)\right)=27-\frac{6}{n}\cdot$$

Thus

$$\lim_{n\to\infty}(U(P_n,f)-L(P_n,f))=\lim_{n\to\infty}\frac{12}{n}=0.$$

Hence f is integrable on $[2,5]$ and

$$\int_2^5 f(x)\,dx=\lim_{n\to\infty}U(P_n,f)=27. \qquad \textbf{8}$$